The V(D)J recombination model and fast generation probabilities in `vdjtools`

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Appendix M — `vdjtools` model engine

Abstract

The `vdjtools` model engine implements the Murugan–Marcou generative model of V(D)J recombination [1, 2] on the AIRR schema [3], storing every model factor as a tidy `polars` table so that the Bayesian-network structure is declared as data rather than hidden in an opaque file format. On top of this representation it computes the generation probability (P_{gen}) of a junction — the total probability that the stochastic recombination process produces a given sequence — both for nucleotide queries (an exact scenario sum) and, more delicately, for amino-acid queries, where one must marginalise over all synonymous nucleotide realisations. For the amino-acid case we use the transfer-matrix factorisation of [4], but re-derive it from a single structural observation — *that the only coupling across a recombination-block boundary is codon translation, never the insertion Markov chain* — which yields a clean left/right partial-sum split (§M.5) that our native C++ core evaluates with machine-precision agreement to `OLGA` on identical legacy parameters (§M.7). The same tables and scenario enumeration drive sequence generation (ancestral sampling) and model inference (expectation–maximisation), which we describe in §M.9.

Scope and terminology. This appendix describes the model engine shipped in `vdjtools` 5.4.0 and distinguishes its working `isalgo` models from frozen `olga-legacy` parameters. An AIRR *junction* includes the conserved Cys104 and Phe/Trp118 codons; the IMGT CDR3 excludes these anchors. The historical OLGA API calls this anchor-inclusive sequence a CDR3. All sequence lengths, coordinates and probabilities below refer to the junction. The notation first introduces the single-D factorisation, then extends it to the joint single/tandem histories used by current VDJ fits. The reference principles are the latent-history likelihood of Murugan et al., including its Supporting Information, and IGoR's probabilistic scenario search [1, 2]; the insertion boundary convention here is stated explicitly below.

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M.1. The generative recombination model

A rearranged receptor chain is the output of a stochastic process that (i) selects germline segments, (ii) trims their ends by exonuclease deletion, and (iii) inserts non-templated nucleotides at the junctions. Following [1, 2] we treat a *recombination scenario* r — the full list of hidden events — as a sample from a Bayesian network whose factors are the model parameters. For a VDJ locus (TRB, TRD, IGH) a scenario is

$$r = (V,\, J,\, D,\, \mathrm{del}_V,\, \mathrm{del}_J,\, \mathrm{del}_D^{5'},\, \mathrm{del}_D^{3'},\, \mathbf{m}^{VD},\, \mathbf{m}^{DJ}), \tag{1}$$

with probability

$$P(r) = P(V)\,P(J)\,P(D \mid J)\;P(\mathrm{del}_V \mid V)\,P(\mathrm{del}_J \mid J)\,P(\mathrm{del}_D^{5'}, \mathrm{del}_D^{3'} \mid D)\;P_{\mathrm{ins}}(\mathbf{m}^{VD})\,P_{\mathrm{ins}}(\mathbf{m}^{DJ}). \tag{2}$$

A VJ locus (TRA, TRG, IGK, IGL) has no D segment; its scenario drops D, the two D-deletions and one insertion, and the gene factor becomes the joint $P(V)\,P(J \mid V)$ with a single VJ insertion. The generated junction nucleotide sequence $\sigma(r)$ is the concatenation, from the conserved Cys 104 to the conserved Phe/Trp 118 inclusive,

$$\sigma(r) = \underbrace{\hat{V}}_{\text{V germline}}\;\underbrace{\mathbf{m}^{VD}}_{\text{N1}}\;\underbrace{\hat{D}}_{\text{D germline}}\;\underbrace{\mathbf{m}^{DJ}}_{\text{N2}}\;\underbrace{\hat{J}}_{\text{J germline}}, \tag{3}$$

where $\hat{V}, \hat{D}, \hat{J}$ are the germline segments after the indicated deletions.

Insertion chains. Each non-templated segment $\mathbf{m} = m_1 \cdots m_\ell$ is drawn from a length distribution and a first-order Markov chain:

$$P_{\mathrm{ins}}(\mathbf{m}) = P_{\mathrm{ins}}(\ell)\; b(m_1) \prod_{i=2}^{\ell} R(m_i \mid m_{i-1}), \tag{4}$$

with b the first-nucleotide distribution and R the 4×4 transition matrix. For zero insertion length, only the length factor contributes. New scaffolds fit b independently, counting the first base of every nonempty insertion, including length-one insertions with no transition. Unobserved transition conditionals retain their normalised initial rows; zeroing them would lose probability mass when first bases and insertion lengths recombine independently. Models without explicit first-base events retain the legacy stationary-R convention. The VD chain is read $5' \to 3'$ and the DJ chain $3' \to 5'$, so b_{DJ} describes the J-proximal insertion nucleotide. This independent first-base distribution differs from IGoR's first transition conditioned on the adjacent retained germline nucleotide. In table form, `from_nt,to_nt,p` stores $R(b \mid a)$ with a the previous nucleotide and b the next; the packed native array uses the equivalent column-stochastic convention $R[b, a]$.

Generation probability. The quantity of interest is the probability of a specified junction, obtained by summing over every scenario that could have produced it,

$$P_{\mathrm{gen}}(\sigma) \;=\; \sum_{r\,:\,\sigma(r)=\sigma} P(r), \qquad P_{\mathrm{gen}}(\alpha) \;=\; \sum_{\sigma\,:\,\tau(\sigma)=\alpha} P_{\mathrm{gen}}(\sigma), \tag{5}$$

where τ is translation: an amino-acid query α additionally marginalises over all synonymous nucleotide sequences. Both sums are finite but astronomically large if taken literally; § M.3 and § M.5 evaluate them by dynamic programming.

M.2. Model parameters as data

`vdjtools` stores a model as a directory of long-format `polars` parquet tables plus a `manifest.json` that names each event and lists its conditioning parents — the edges of the Bayesian network of Eq. (2). The engine implements the declared supported factorisations; changing parents alone does not implement a richer dependency such as $P(D \mid V, J)$, which also requires compatible packing, inference and sampling. A VJ locus omits the D tables and second insertion. Each table holds the distribution of one event, normalised within its conditioning key by a single `polars group_by` — which is exactly the M-step of § M.9. Two conventions matter for the P_{gen} recursions.

P-nucleotides. A negative deletion count encodes templated palindromic (P-) nucleotides. Each germline cut segment is pre-extended by up to `maxpal` reverse-complement nucleotides; a "deletion" of $-k$ means k P-nucleotides are retained. The stored count del is thus *biological* (negative = P-nucleotides), while the dense array index used at run time is del + `maxpal` ≥ 0.

Germline references and gene resolution. Native scaffolds resolve V/D/J sequences and anchor coordinates from `arda`. V germline is cut from its anchor to its 3′ end; J germline extends from its 5′ end through the anchor codon. Anchor offsets are zero-based nucleotide coordinates in each full reference, not codon indices; references can differ in framework length or sequence even when these conventions agree. A model never mixes germline sources: models imported from `OLGA` retain their imported reference, whereas native fits use `arda`. Raw-OLGA engine comparisons explicitly disable anchor repairs so both engines receive identical germline sequences. New native models select one usable, complete, functional major allele per gene: `*01` when available, otherwise the lowest-numbered usable allele. Allele suffix aliases resolve to this representative without adding probability mass; ambiguity between different genes remains in the scenario sum. Explicit all-allele models retain allele-specific templates and scoring. Unknown calls raise rather than silently becoming unrestricted gene sums.

The D-D extension. The schema carries single- and tandem-D rearrangements, including a second-D deletion and a DD insertion, beyond the single-D event graphs used here as `OLGA`/IGoR references (§ M.4). The current joint-history models use $n_D \in \{1, 2\}$; a fully deleted single D remains a one-D latent history, even though it contributes zero retained nucleotides. A strict single-D model has no tandem history by construction.

M.3. Nucleotide generation probability

For a nucleotide query σ the inner sum of Eq. (5) is a direct enumeration, because each candidate germline placement fixes the remaining insertion strings. Writing $N = |\sigma|$ and letting a V (resp. J) contribute $\ell_V \geq 1$ (resp. $\ell_J \geq 1$) nucleotides — a valid rearrangement is never fully V- or J-deleted, matching `OLGA` —

$$P_{\text{gen}}(\sigma) = \sum_{V,J} G(V,J) \sum_{\substack{\ell_V,\ell_J \geq 1,\ \ell_V+\ell_J \leq N \\ \sigma[0:\ell_V]=\hat{V},\ \sigma[N-\ell_J:]=\hat{J}}} P(\text{del}_V \mid V) P(\text{del}_J \mid J)\ \Xi_J(\sigma[\ell_V : N-\ell_J]), \qquad (6)$$

where $G(V,J) = P(V)P(J \mid V)$ for VJ and $G(V,J) = P(V)P(J)$ for VDJ. For a VJ locus $\Xi_J(\mu) = P_{\text{ins}}(\mu)$ is the single-junction insertion weight of Eq. (4), and for a VDJ locus Ξ_J is itself a sum over the D gene, its two deletions and its placement,

$$\Xi_J(\mu) = \sum_D P(D \mid J) \sum_{\text{del}_D^{5'},\text{del}_D^{3'}} P(\text{del}_D^{5'}, \text{del}_D^{3'} \mid D) \sum_{\substack{\text{placements } p \\ \mu[p:p+\ell_D]=\hat{D}}} P_{\text{ins}}(\mu[0{:}p])\, P_{\text{ins}}(\mu[p+\ell_D{:}]). \tag{7}$$

Common-prefix / common-suffix matching restricts ℓ_V, ℓ_J to the genes that can actually flank σ, so Eq. (6) is fast in practice. This exact nucleotide sum supplies the zero-error E-step; the observation-error channel modifies its germline emission factors (§ M.9). On identical raw legacy parameters, the native port reproduces `OLGA`'s `compute_nt_CDR3_pgen` to machine precision.

M.4. Tandem-D (D-D) rearrangements

A tandem-D locus joins two D segments in the junction, $V\text{—}D_1\text{—}D_2\text{—}J$. New joint-history models store

$$q_J(n, D_1, D_2) = P(n, D_1, D_2 \mid J), \qquad \sum_{(n,D_1,D_2)\in\mathcal{S}_J} q_J(n, D_1, D_2) = 1. \tag{8}$$

Here $\mathcal{S}_J$ contains physically permitted histories, $n \in \{1, 2\}$, and D_2 is absent for $n = 1$. Thus the single-D probability can vary with J; there is no independent global $P(n_D)$ factor in this representation. TRB permits D_1–D_2–J_2 and single D_1–J_2 histories. Human TRD histories follow the curated forward coding order TRDD1–TRDD2–TRDD3. Constraints act on complete histories before fitting, not on fitted probability tables afterwards. The implemented extension retains at most two D genes; it does not enumerate three-D histories. No universal 10% tandem-D ceiling is imposed. A preliminary greedy alignment cannot veto a permitted history.

A two-D scenario has probability

$$\begin{aligned} P(r_2) = {} & P(V)P(J)q_J(2, D_1, D_2)P(\text{del}_V \mid V)P(\text{del}_J \mid J) \\ & \times P(\text{del}_{D_1} \mid D_1, n = 2)P(\text{del}_{D_2} \mid D_2, n = 2) \prod_{u\in\{VD,DD,DJ\}} P_{\text{ins}}(\mathbf{m}^u). \end{aligned} \tag{9}$$

Its junction is $\sigma(r_2) = \hat{V}\, \mathbf{m}^{VD}\, \hat{D}_1\, \mathbf{m}^{DD}\, \hat{D}_2\, \mathbf{m}^{DJ}\, \hat{J}$. For $n = 1$, use $q_J(1, D_1, \varnothing)$, its separate single-D deletion conditional, and omit the D_2 and DD factors. Consequently

$$\Xi_J(\mu) = \sum_{r:\, \sigma_{\text{middle}}(r)=\mu} q_J(n, D_1, D_2)P(\text{middle trims and insertions} \mid n, D_1, D_2). \tag{10}$$

Proposition 1 (Disjoint latent histories). *Both retained D segments in a tandem history have at least one nucleotide. Single-D histories allow complete D deletion. These are disjoint sets of latent histories, but their emitted nucleotide sequences can overlap. Their mixture is normalised when q_J and every reachable conditional are normalised.*

Requiring nonempty tandem segments is an explicit model convention; it does not make the number of D segments identifiable from a sequence. In particular, insertions can coincidentally match

germline bases. The inference and generation kernels use separate normalised single-D, tandem-first-D and second-D deletion tables, so removing fully deleted tandem states does not leak probability. EM accumulates posterior counts directly over $\mathcal{S}_J$, then normalizes within J. An optional single-D pseudocount is divided uniformly among declared single-D history rows within each J. Undefined groups retain their initial normalised conditional. The reported D-count probability is $P(n) = \sum_J P(J) \sum_{D_1,D_2} q_J(n, D_1, D_2)$.

Native and reference nucleotide/amino-acid Pgen, generation and E-step kernels use these same factors. The native codon-constrained max-product search also includes tandem paths and reconstructs both D spans. Legacy models retain their existing global $P(n_D)P(D_1 \mid J)P(D_2 \mid D_1)$ representation and artefact behavior; the joint-history schema is an explicit conversion, not a reinterpretation of frozen tables.

M.5. Amino-acid generation probability by transfer matrix

The amino-acid case is harder: the codon marginalization in Eq. (5) reintroduces a sum over the unobserved junction nucleotides. Enumerating scenarios and summing synonymous codons position by position (a codon-constrained dynamic program of width $|\Sigma|$ over the nucleotide state) is correct but costs one $O(N)$ pass per $(D, \mathrm{del}_D^{5'}, \mathrm{del}_D^{3'}, \text{placement})$ skeleton — quadratic in N and prohibitive for long junctions. The transfer-matrix method reuses partial sums across placements and insertion lengths.

Setup. Let $\alpha = \alpha_0 \cdots \alpha_{L-1}$ be the query, $N = 3L$, and σ_i the (summed over) nucleotide at position i. Codon c imposes $\tau(\sigma_{3c}, \sigma_{3c+1}, \sigma_{3c+2}) = \alpha_c$. Because the codon and the insertion Markov chain each look back at most two and one nucleotide respectively, a left-to-right dynamic program with state $x_i = (\sigma_{i-1}, \sigma_{i-2})$ suffices: it can (a) close a codon when $i \equiv 2 \pmod 3$ and (b) carry the Markov dependency. The width is $|\Sigma| = 25$ (four nucleotides plus a start sentinel, squared). This is the codon-marginalizing DP; the transfer matrix is what lets us reuse its partial results across all scenarios.

The key structural fact. In the block layout of Eq. (3), $\hat{V}$–$\mathbf{m}^{VD}$–$\hat{D}$–$\mathbf{m}^{DJ}$–$\hat{J}$, the blocks are coupled only through codon translation, never through an insertion Markov chain.

Proposition 2 (Codon-only coupling). *For fixed gene choices and trimming events, the nucleotide-dependent weight across each block boundary in Eq.* (3) *factors into a left factor, the indicator(s) of the codon(s) straddling the boundary, and a right factor. No transition $R(\cdot \mid \cdot)$ spans a boundary.*

Sketch. The VD chain's leftmost nucleotide m_1^{VD} is drawn from the bias b_{VD} independent of the last V nucleotide, and its rightmost nucleotide carries no transition into $\hat{D}$; the D block is germline (fixed nucleotides, no chain); the DJ chain, read $3' \to 5'$, draws its J-proximal nucleotide from b_{DJ} independent of the first J nucleotide, and its D-proximal nucleotide carries no transition into $\hat{D}$. Hence every factor in Eq. (4) lives strictly inside one block. The only inter-block constraints are the codon indicators $\mathbb{1}[\tau(\cdots) = \alpha_c]$ for codons that straddle a boundary. □

Proposition 2 licenses computing the left and right halves of the sum *once each* and stitching them, instead of recomputing a full pass per scenario.

Left partial sums. Define, for each boundary position p (the position at which $\hat{D}$ starts) and each nucleotide state $x = (\sigma_{p-1}, \sigma_{p-2})$,

$$\Lambda_p(x) = \sum_{\substack{V,\,\mathrm{del}_V,\,\ell_{VD} \\ \ell_V + \ell_{VD} = p}} P(V) P(\mathrm{del}_V \mid V)\, P_{\mathrm{ins}}(\ell_{VD})\, [\,VD \text{ Markov}\,] \prod_{\text{codons} \subset [0,p)} \mathbb{1}[\tau = \alpha], \tag{11}$$

the total weight of all ways the V germline (with $\ell_V \geq 1$) plus the VD insertion fill $[0, p)$ ending in state x, with every codon that lies wholly in $[0, p)$ satisfied. Λ is built by one forward recursion: germline seeds are deposited at $p = \ell_V$, then the VD transfer step — multiply by b_{VD} on the first inserted nucleotide and R_{VD} thereafter, close codons at $p - 1 \equiv 2 \pmod 3$, and add $P_{\mathrm{ins}}(\ell_{VD})$ at each length — extends them, in $O(NK_{VD}|\Sigma|)$ work after seeding, where K_{VD} is the maximum supported VD insertion length, truncated to N. The code starts a separate insertion recursion from each supported germline boundary; it is quadratic in N when K_{VD} grows with N.

For a seed boundary q, let $W_{q,0}$ be its germline-seeded state vector, including gene and deletion weights. For $\ell \geq 1$, the forward update is

$$W_{q,\ell}(a, b) = \sum_c W_{q,\ell-1}(b, c) \begin{cases} b_{VD}(a), & \ell = 1, \\ R_{VD}(a \mid b), & \ell > 1 \end{cases} \chi_{q+\ell-1}(c, b, a), \tag{12}$$

where a, b, c are nucleotide states and χ_i is one unless $i \equiv 2 \pmod 3$; in that phase it tests the completed codon against the query. Unavailable leading bases are represented by the sentinel and never enter a completed codon test. Then $\Lambda_p = \sum_{q+\ell=p} P_{\mathrm{ins}}(\ell) W_{q,\ell}$, including $\ell = 0$. Insertion length weights enter this sum, not the propagation from ℓ to $\ell + 1$.

Right partial sums. Symmetrically, for each D gene and boundary p (where $\hat{D}$ ends),

$$\Psi^D_p(y) = \sum_{J,\,\mathrm{del}_J,\,\ell_{DJ}} P(D \mid J) P(J) P(\mathrm{del}_J \mid J)\, P_{\mathrm{ins}}(\ell_{DJ})\, [\,DJ \text{ Markov}\,] \prod_{\text{codons} \subset [p,N)} \mathbb{1}[\tau = \alpha], \tag{13}$$

with $y = (\sigma_p, \sigma_{p+1})$ and $p + \ell_{DJ} + \ell_J = N$ in the sum. Because the DJ chain runs $3' \to 5'$, Ψ^D is built by a *backward* recursion from the J boundary — one per D gene, the $P(D \mid J)$ weight folded into the seed. There are at most d such passes, where d is the number of D templates, each costing $O(NK_{DJ}|\Sigma|)$ after seeding. This d is distinct from the latent D-count event n_D.

Stitching. Fixing D, its deletions (hence the fixed germline string $\hat{D}$ of length ℓ_D) and its start p_0, thread the germline nucleotides of $\hat{D}$ through the DP from Λ_{p_0} to a state distribution at $p_1 = p_0 + \ell_D$ (closing any codons interior to $\hat{D}$, no Markov weight), then contract with $\Psi^D_{p_1}$. By Proposition 2 at most one codon straddles the D/DJ boundary p_1; the contraction is a bounded codon-state contraction (or scalar product when $p_1 \equiv 0 \pmod 3$) over that codon's free nucleotides. Summing over D, its deletions and p_0,

$$P_{\mathrm{gen}}(\alpha) = \sum_D \sum_{\mathrm{del}^{5'}_D, \mathrm{del}^{3'}_D} P(\mathrm{del}^{5'}_D, \mathrm{del}^{3'}_D \mid D) \sum_{p_0} \left\langle \mathcal{T}_{\hat{D}}\, \Lambda_{p_0}\,,\ \Psi^D_{p_1} \right\rangle_{p_1}, \tag{14}$$

where $\mathcal{T}_{\hat{D}}$ is the germline-threading operator and $\langle \cdot, \cdot \rangle_{p_1}$ the codon-contracting stitch at p_1. The insertion-length sums are already inside Λ and Ψ^D, so the outer work is $O(d \cdot \#\mathrm{del}_D \cdot N \cdot \ell_D)$ with

$\ell_D \ll N$ — versus the enumeration's extra factor of N from re-running the DP per placement. For a VJ locus there is no D and no D bridge: for each retained J template, the code builds a left DP with $P(J \mid V)$ in the V seed and threads each supported J suffix into its boundary state. The resulting computation still sums all supported V/J trims and insertion lengths.

Remark 1. This is the $\Pi_L \cdot \Pi_R$ factorisation of [4] expressed on the `polars` tables. Where `OLGA` handles the D block with a sliding right-deletion recursion folded into Π_R, we keep Ψ^D purely J/DJ and slide a short explicit germline bridge (Eq. (14)); the two agree to machine precision (§ M.7), and the explicit bridge is what generalizes to the tandem-D event.

Exact tandem contraction. Let N be the junction nucleotide length, $S = 25$ the codon-state count, D the number of functional D templates, and J the number of selected J alleles. The native engine builds the V/VD left matrix once. For each first D template d, it sums all supported nonempty retained-D histories ending at each boundary/state and extends that sum through the DD insertion recurrence. Call the resulting matrix M_d. This matrix contains the first-D deletion weight but no J-dependent history weight. For each J allele j, R_j contains $P(J = j)$, J trimming and DJ insertion factors. Let H_{jde} denote the joint tandem-history weight for first D d and second D e, or the legacy product $P(d \mid j)P(e \mid d)$. Forbidden complete histories have $H_{jde} = 0$.

For fixed second D e, let $C_e(A, B)$ mean threading every supported nonempty trim of e through left message A, multiplying by its second-D deletion probability, and stitching with right message B using Eq. (14). This operation is bilinear, so the tandem branch is

$$T = \sum_{d,e} C_e\left(M_d, \sum_j H_{jde} R_j\right) = \sum_{j,e} C_e\left(\sum_d H_{jde} M_d, R_j\right). \tag{15}$$

The global tandem-mixture weight, when present, multiplies T exactly once in the caller. The implementation chooses the first expression when $D \leq J$ and the second otherwise. Thus it performs at most $\min(D^2, JD)$ second-D bridge walks, rather than JD^2, and computes each M_d only once. Mixing messages costs $O(JD^2NS)$ in the dense worst case; zero history weights are skipped. Message storage is $O((J+D)NS)$ in the second form and $O(JNS)$ in the first, excluding the existing fixed-size codon-state scratch.

Both D roles share a prefix walk over placements and 5-prime trims. Successive retained lengths extend the same prefix instead of rethreading each 3-prime trim from its beginning. A zero codon-compatible message terminates every longer prefix exactly. After two fixed bases, the message has only one possible codon state, so subsequent fixed bases use a scalar state transition. The first- and second-base messages are shared across 5-prime cuts for each input matrix. At a DD-insertion end, completed codons can be forgotten before the fixed second-D block: boundary phases zero, one and two need respectively zero, one and two past bases. Thus at most 1, 4 or 16 canonical states remain, further restricted by the codon mask. Only those cells enter the D-pair mixing loop. The insertion's internal Markov recurrence retains its full state; forgetting its last inserted base before the insertion ends would be incorrect. All nonzero deletion weights remain; a prefix shorter than the first supported length still propagates because it can lead to a longer supported fragment. Each D retains at least one nucleotide and J retains at least one nucleotide. No minimum retained-D length beyond this model convention, probability cutoff, or renormalization is used. Linearity proves equivalence in

exact arithmetic; changing nonnegative reduction order can change floating-point rounding. Native singleton-codon NT masks, AA masks, degenerate masks and Hamming-1 calls use this same kernel. The V+VD message is computed once and shared across single-D and tandem branches of the mixture; single-D coefficients and frozen OLGA probabilities retain their original meanings.

For each right message, the boundary join has an adjoint coefficient vector $B_j(q,s)$ such that $\mathrm{join}(A,R_j;q)=\sum_s A(q,s)B_j(q,s)$. It is computed once per right message and reused over all second-D trims. After two fixed D bases only one left state survives, so each terminal join becomes one multiplication. This is the same nonnegative bilinear form; no retained-prefix contribution is omitted.

For a nonnegative sum of k fixed floating-point terms, sequential accumulation has relative rounding-error bound $\gamma_k = ku/(1-ku)$ when $ku<1$, where u is unit roundoff. This bound concerns the sum alone, not errors already present in its terms. A contraction with fewer additions can therefore be more accurate than the original enumeration while differing from its double result. In the Linux 1,000-query IGH comparison, 32 differences exceeded 10^{-12} relative to the old double accumulation. Re-evaluating all 32 with the original kernel and only its final tandem accumulator promoted to long double gave maximum relative errors 4.8×10^{-13} for the optimized contraction and 5.9×10^{-12} for the original double result. No probability cutoff was introduced to obtain this agreement.

M.6. Optional coarse post-selection calibration

The recombination model and its error-free generation probabilities remain unchanged. The optional selection model retains a joint V/J factor and a shared length factor from the factorized approach of Elhanati et al. [5], without amino-acid-position factors. Here L counts amino acids including the conserved C and terminal F/W. Let $g=(v,j)$ be a gene pair and let $P_0(g,L)$ be the exact recombination mass conditioned on a C-first, F/W-last, stop-free junction, computed by degenerate-AA DP. Its unconditioned eligible mass E_0 is retained separately. Define

$$S(g,L)=\frac{\exp(a_g+b_L)}{Z},\qquad Z=\sum_{g,L}P_0(g,L)\exp(a_g+b_L). \tag{16}$$

Thus $\sum P_0 S=1$; the length-only option fixes $a_g=0$. For observation stratum i, let y_i count independent receptor events, L_i be its anchored length and C_i the compatible gene-pair set. Missing calls expand this set; preliminary calls incompatible with every model cell at that length are reconsidered over all pairs. Truly unsupported lengths are audited and excluded. With $n=\sum_i y_i$ over supported strata, maximize

$$\sum_i y_i \log\left[\sum_{g\in C_i} P_0(g,L_i)e^{a_g+b_{L_i}}\right] - n\log Z - \frac{\rho}{2}(\|a\|^2+\|b\|^2) - \frac{\tau}{2}\|\Delta^2 b\|^2. \tag{17}$$

Here $\rho>0$ fixes the log-weight gauges and shrinks sparse cells; $\tau\geq 0$ smooths the shared log length curve through second differences. Penalties use event-count likelihood units. Ambiguous observations use the sum inside the logarithm, rather than a fixed choice of one gene. Reports retain optimizer diagnostics and unsupported/reassigned event counts.

A separate effective multiplicity scale Q is fitted after the shape. For an observed AA junction σ_i, the model mass entering this fit is

$$p_i = \sum_g P_{\text{gen}}(\sigma_i, g) S(g, |\sigma_i|), \qquad \lambda_i = NQp_i, \tag{18}$$

where N is the sampled eligible receptor-event exposure. The observed count y_i follows a zero-truncated Poisson likelihood; the positive maximum satisfies $\sum_i [y_i - \lambda_i/(1 - e^{-\lambda_i})] = 0$. Zero-mass junctions and their events are reported separately, never counted as scored observations. This effective scale is not a measured thymic survival probability. Gene-dependent weights enter the latent gene sum; multiplying marginal Pgen by one chosen gene factor would be incorrect. A complete parameter hash binds the fitted shape to its recombination model. Event subsampling precedes V/J/AA reduction and samples integer receptor events without replacement, rather than read abundance.

In ALICE's fixed-V/J substitution ball, length is constant, so $S(g, L)$ can multiply the eligible joint ball once. Profiled balls retain C-first/F/W-last neighbors. The reference-panel consumer applies S before its AA-maximum reducer, then corrects residual within-V J composition using $\sum_L P_0(g, L) S(g, L)$ rather than the unselected gene marginal.

M.7. Correctness checks and recorded benchmark

The native transfer matrix is validated against two independent oracles: the transparent amino-acid scenario enumeration of § M.5 (pure Python), and `OLGA` itself [4]. On identical imported human TRB parameters it reproduces the published worked example

$$P_{\text{gen}}(\texttt{CAWSVAPDRGGYTF}, \texttt{TRBV30*01}, \texttt{TRBJ1-2*01}) = 1.2036 \times 10^{-10}.$$

It agrees with `OLGA` to a maximum relative error of 1.2×10^{-15} over 25 marginalised (all V/J/D) and 6 V/J-restricted generated junctions — i.e. to machine precision. Table 1 reports historical timing, whose producer now resides in the separate benchmark project as `bench/bench_pgen.py` on the human TRB `OLGA` default model, single-threaded, over 20 marginalised queries.

Table 1: Recorded amino-acid P_{gen} throughput on human TRB (single thread, marginalised over all V/J/D). TM is the Eq. (14) transfer matrix in the native `_core`; OLGA is `compute_aa_CDR3_pgen`. These are historical measurements on 20 queries with the frozen single-D OLGA parameter family, not timings of the current isalgo fits. Lower is better.

Method	ms / sequence	vs OLGA	max. rel. error vs OLGA
OLGA	5.3	1.0×	—
TM (native `_core`)	**0.64**	**8.3**×	1.2×10^{-15}

The pure-Python scenario enumeration of § M.5 is exact and serves as the correctness oracle, but is not a runtime path: unrestricted, it costs on the order of minutes per query (its extra factor of N and its lack of the left/right reuse), which is exactly the cost the transfer matrix removes.

M.8. Degenerate, agnostic and Hamming-ball queries

The same transfer matrix supports three additional query types. First, the codon test at each amino-acid position is a *membership test against an allowed-codon set* (a 64-bit mask), not an equality against one residue; a wildcard mask (any of the 20 amino acids) or any degenerate set is therefore evaluated in a single pass, giving fixed-length motif P_{gen} without enumerating the matching sequences. Second, conditioning on a V and/or J allele is a usage mask over the gene sums of Eqs. (11), (13); the empty restriction marginalises over all functional genes. These restricted sums are joint sequence-and-gene probabilities, not renormalised probabilities conditional on observing a gene. An unrestricted query is *V-agnostic / J-agnostic* P_{gen}.

Combining the two gives a fast one-mismatch query. The total P_{gen} of an amino-acid junction and every sequence within Hamming distance 1 (one substitution) is, by inclusion–exclusion,

$$P_{\text{gen}}^{\leq 1}(\alpha) = \sum_{k=0}^{L-1} P_{\text{gen}}(\alpha_{k\to *}) - (L-1)\, P_{\text{gen}}(\alpha), \tag{19}$$

where $\alpha_{k\to *}$ wildcards position k. `OLGA` uses the same identity but pays a full (slow) transfer-matrix pass for each of the $L+1$ terms; here each term is one fast pass (Eq. (14)) with a single wildcard mask, so the native `_core` evaluates Eq. (19), with recorded historical throughput of $8.7\times$ `OLGA`'s throughput on human TRB (67.9 vs 7.8 ms/query) and $2.5\times$ on human TRA (25.1 vs 10.1 ms/query), matching it to a relative error of 10^{-15} on the recorded legacy comparison. These timings do not establish current isalgo throughput.

M.9. Generation and inference on the same tables

Generation. Ancestral sampling walks the Bayesian network of Eq. (2) top-down: draw $(V, J[, D])$, the deletions, the insertion lengths, then the insertion nucleotides from the Markov chain, and assemble Eq. (3). Restricting to functional genes and enforcing $\ell_V, \ell_J \geq 1$ keeps emitted junctions within the scorer's structural support. The sampler also enforces disjoint D histories and genomic order. The library primitive retains its reference sampler by default; the generation CLI selects native batched sampling for bundled isalgo models. Native draws are reproducible across thread counts for a fixed seed but use a different random stream from the reference sampler. Productive-only sampling rejects out-of-frame or stop-bearing draws, so its observed marginals are conditional on that filter rather than the unfiltered model.

Current model catalogue. Twelve bundled isalgo fits cover all seven human loci and mouse TRA, TRB, IGH, IGK and IGL; mouse TRG/TRD are not bundled. The old `learned` and `arda` source names alias isalgo; frozen OLGA parameters are selected explicitly as `olga-legacy`. Generation and scoring use the selected model's reference and factors without substituting older weights for a missing fit.

Inference. We estimate model parameters from nonfunctional reads using EM. The raw-data builder initialises the retained-germline substitution rate at 0.01 and fits it; the lower-level `infer_native` API defaults to `error_rate=None` for exact matching. Exact-match inference uses the scenario sum of § M.3. With the optional retained-germline substitution channel, let m_r be the number

of mismatches and g_r the number of retained germline bases in scenario r for observed sequence y. Its joint weight and observation likelihood are

$$w_r = P_\theta(r)(\epsilon/3)^{m_r}(1-\epsilon)^{g_r-m_r}, \qquad L(y) = \sum_r w_r, \qquad P(r \mid y) = w_r/L(y). \tag{20}$$

Here ϵ is the substitution probability, shared across retained germline bases; each alternative nucleotide has probability $\epsilon/3$. Inserted bases retain their observed Markov emissions without an observation-error channel. This scope differs from applying sequencing errors to every generated base. Exact generation P_{gen} remains the error-free quantity of § M.3. Sequencing observations are normalised to A/C/G/T before alignment and inference: DNA IUPAC ambiguity symbols are replaced by A. They are low-quality input symbols, not latent nucleotide alternatives, so inference does not enumerate or marginalise compatible completions. This input rule does not alter germline references or amino-acid patterns. The substitution channel above applies to ordinary A/C/G/T mismatches in the normalised observation. Invalid non-IUPAC DNA symbols raise an error; normalisation occurs on the original strand before reverse complementation. The error-rate M-step, for equally weighted input observations, is

$$\epsilon_{\text{new}} = \frac{\sum_y E[m_r \mid y]}{\sum_y E[g_r \mid y]}. \tag{21}$$

For tandem histories, the forward/backward messages include the emission factors for each retained D segment. Posterior mismatch and retained-length statistics are attributed in the reconstruction passes, once per segment; the outer V/J statistics are weighted by the complete history mass. Event tables are normalised within their conditioning keys using the same posterior weights.

ARDA annotations provide preliminary gene proposals, nonfunctional labels and alignment evidence, not fixed gene or allele assignments. The raw-data builder obtains V/J proposals in one batched multi-mapping search; proposed genes expand to the model's functional allele templates, while missing proposals leave that segment unrestricted. D and D2 remain unrestricted by preliminary annotations. The native error-aware frame API uses the same proposal route; exact-mode call masks remain an explicit restriction of its input support. Preliminary alignment endpoints do not fix recombination boundaries. Exact inference retains all supported shorter retained segments and corresponding N/P histories in the sum; coincidentally matching N bases do not imply zero insertion. Restricting enumeration to a candidate gene set is a computational approximation that can change the likelihood and fitted parameters if compatible alternatives are omitted. It is not an invariance of the EM fixed point. Recorded positive-likelihood observation counts exclude unsupported inputs. A mean log-likelihood over that subset must not be interpreted as a finite full-cohort likelihood when any observation has zero support.

Inference flow. The implementation follows this sequence:

```
normalise observations; select reference and permitted genomic histories
build batched V/J proposals; retain D/D2 ambiguity
repeat until tolerance or iteration cap:
    sum full and retained history weights for each supported observation
    accumulate retained posterior event, mismatch and length counts
```

```
normalise event tables; update substitution rate
report likelihood, supported/unsupported counts and omitted mass
```

The default raw-data build uses at most 15 iterations and relative likelihood change tolerance 10^{-4}; the lower-level native API defaults to 30 iterations and 10^{-3}. These are convergence settings, not biological thresholds.

Bounded native inference. The working inference default is a relative scenario cutoff $\rho = 10^{-5}$; $\rho = 0$ retains exact inference. Let $\mathcal{H}$ partition complete scenarios into disjoint cells: a single-D gene/deletion/placement cell, a tandem-D bridge, or a complete VJ cell. Max-product messages give

$$M_C = \max_{r \in C} w_r, \qquad M = \max_{C \in \mathcal{H}} M_C, \qquad \mathcal{K}_\rho = \{C \in \mathcal{H} : M_C \geq \rho M\}. \tag{22}$$

For $L(y) > 0$, the retained likelihood and omitted probability fraction are

$$L_\rho(y) = \sum_{C \in \mathcal{K}_\rho} \sum_{r \in C} w_r, \qquad \delta_\rho(y) = 1 - \frac{L_\rho(y)}{L(y)}. \tag{23}$$

Every retained event/error count uses that same cell set and normalisation $L_\rho(y)$. Some weak component histories remain within retained cells; this preserves factorisation. The relative bound follows IGoR's inference principle, but these global-max groups differ from its dynamic traversal cutoff. The cutoff does not bound total omitted mass: its mean and maximum over scoreable observations are recorded per iteration. Generation P_{gen} remains exact, and no fixed reassignment-length cap is used. Exact EM with fixed support and unregularised M-steps has the usual nondecreasing training-likelihood property; a changing retained cell set or pseudocount-regularised updates do not establish that guarantee for the exact unpenalised objective. A training-likelihood increase is not independent validation.

M.10. Model diagnostics: entropy and mutual information

Because the Bayesian network is declared as data (§ M.2), every model is directly introspectable: each event table is a (conditional) distribution, so we quantify each part of the rearrangement by its Shannon entropy and each network edge by mutual information, straight from the tables and with no reference to any sequence. For an event X with parents $\mathrm{Pa}(X)$ we report the marginal entropy $H(X)$ (in bits) and, along each declared edge, the information it carries,

$$I(X; \mathrm{Pa}(X)) = H(X) - H(X \mid \mathrm{Pa}(X)), \qquad H(X \mid \mathrm{Pa}) = \sum_{\mathrm{pa}} P(\mathrm{pa})\, H(X \mid \mathrm{pa}), \tag{24}$$

where parent marginals are obtained by forward propagation over the network. If some parent states have an undefined (all-zero) conditional, these diagnostics condition on the remaining defined support: with $A = \{\mathrm{pa} : \sum_x P(x \mid \mathrm{pa}) > 0\}$ and $Z = \sum_{\mathrm{pa} \in A} P(\mathrm{pa})$, both the child marginal and the conditional entropy use weights $P(\mathrm{pa})/Z$ on A. The same convention applies to within-D deletion coupling. Entirely undefined events ($Z = 0$) report zero marginal mass and zero entropy; no uniform mass is imputed. These are summaries on defined event support, including in the scenario-entropy table. Rendering the manifest as a graph, with nodes annotated by $H(X)$ and supported edges by I, gives

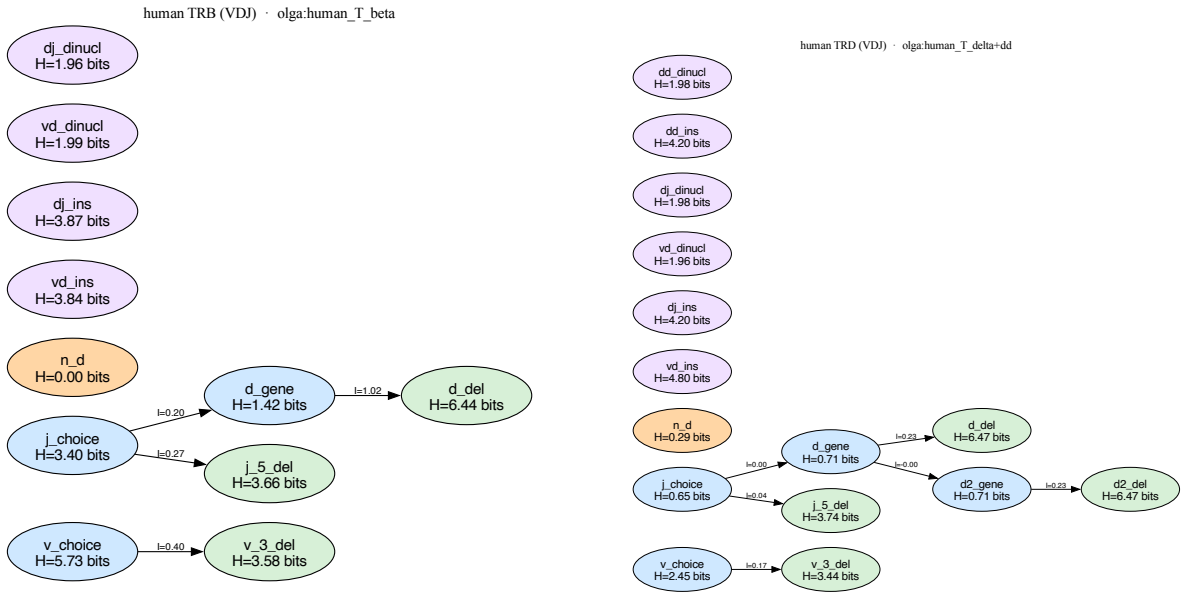


Figure 1: Historical recombination Bayesian networks rendered from frozen manifests, nodes labelled with marginal entropy H (bits) and edges with mutual information I (bits). *Left:* human TRB (single-D). *Right:* human TRD with the legacy tandem-D representation, adding the n_D, D_2-gene, D_2-deletion and DD-insertion nodes. Current schema-3 fits replace the separate D-choice/count factors with $q_J(n, D_1, D_2)$ and separate branch-specific deletion tables (§ M.4).

a `bnlearn`-style summary of the whole model at a glance. Figure 1 retains two historical network renderings.

Table 2 retains a historical comparison of the per-event entropy across five frozen OLGA-reference loci; each row's most entropic locus is in bold. Gene choice dominates in the $\alpha\beta$ loci ($H_V \approx 5.7$ bits for TRB, $H_J \approx 5.6$ for TRA), whereas TRD spends little on gene choice ($H_V = 2.4$ bits, few δ genes) and the most on non-templated insertion ($H_{VD} = 4.8$ bits) — the TdT signature. The single most entropic event in every VDJ locus is the joint D-deletion (up to 7.8 bits for IGH), and its two ends are far from independent: the within-D coupling $I(\mathrm{del}_D^{5'}; \mathrm{del}_D^{3'} \mid D) = 1.18$ bits for TRB (a genuine conditional mutual information, averaged over D so that the shared choice of D does not inflate it). Two edges are worth naming. First, $I(V;J) = 0$ for every VDJ locus is not a measurement but an *assumption* made visible — `OLGA` factors $P(V,J) = P(V)P(J)$ — whereas the VJ loci carry $I(V;J) > 0$ through $P(J \mid V)$ (e.g. 0.27 bits for TRA). Second, the D-deletion edge $I(\{\mathrm{del}_D\}; D)$ is large (1.02 bits for TRB), i.e. trimming depends strongly on which D was chosen. The identical routine applies to an EM-inferred model, so the same table compares a model learned from real reads against the `OLGA` bootstrap parameter-by-parameter. An earlier single-D, aligned-call-masked diagnostic on 6000 out-of-frame human TRB clonotypes reported D-deletion entropy changing from 6.44 to 7.65 bits and VD-insertion entropy from 3.84 to 4.50 bits, with within-D deletion coupling near 1 bit under both fits. Those values describe that diagnostic

Table 2: Marginal entropy $H(X)$ (bits) of each recombination event across five human loci, computed from frozen OLGA-reference model tables. These values are retained as historical diagnostics, not summaries of the current isalgo fits. Dashes mark events absent from a locus (VJ loci have no D; VDJ loci have no VJ insertion). Bold marks the most entropic locus per event.

Event	TRB	TRA	TRD	IGH	IGK
V choice	**5.73**	5.37	2.45	5.71	4.81
J choice	3.40	**5.57**	0.65	2.17	2.83
D choice	1.42	—	0.71	**4.30**	—
V 3′ deletion	3.58	**3.69**	3.44	3.46	3.15
J 5′ deletion	3.66	3.89	3.74	**4.13**	3.65
D deletion (5′,3′)	6.44	—	6.47	**7.78**	—
VD insertion length	3.84	—	**4.80**	4.68	—
DJ insertion length	3.87	—	4.20	**4.82**	—
VJ insertion length	—	**3.84**	—	—	3.59
VD dinucleotide	**1.99**	—	1.96	1.98	—
DJ dinucleotide	1.96	—	1.98	**1.99**	—
VJ dinucleotide	—	**1.97**	—	—	1.92

fit, not the current error-aware, ambiguity-preserving isalgo models. The producer's "held-out" label referred to its own training-report likelihood; it did not constitute a held-out evaluation. The independent functional-source validator in `compare_models.py` is described below and has a different purpose.

A separate historical annotation summary reported two aligned D segments in 4.15% of out-of-frame and 3.42% of in-frame unique human TRD clonotypes. These are preliminary annotation fractions, not estimates of the current model's posterior tandem frequency; the original summary did not record the denominators.

M.11. Implementation, validation and scope

Code map. The Python reference scenario sums are in `model/pgen.py`; packed native scoring and the transfer matrices are in `src/pgen.cpp`, exposed by `model/native.py`. `model/dd.py` declares and validates joint-history support; `model/infer.py` owns proposals, EM updates and reports; `model/data.py` owns the raw-data build. `model/generate.py` provides reference/native sampling, and `model/analyze.py` provides entropy and mutual-information summaries. Paths beginning with `model/` are relative to `python/vdjtools/`.

Exactness and resources. The probability sums are exact over the selected model support, apart from floating-point arithmetic; "exact" does not imply that the reference includes every biological history. Zero-mass branches are skipped algebraically without approximation. The native single-D implementation shares short D-prefix states and reduces right-boundary states within each query; these are transient DP workspaces. With $S = 25$ codon states, d D templates, T supported D deletion pairs per template, maximum retained D length G, and insertion bounds K_{VD}, K_{DJ}, a conservative

bridge bound is $O(dTNGS)$, in addition to germline seeding and $O(NS(K_{VD} + dK_{DJ}))$ insertion recursions. The single-D query workspace is $O(NS + NG)$ in addition to packed model tables. Tandem scoring adds a J/D-pair traversal and a DD-insertion bridge; it is not covered by a universal linear-in-length bound. Native batch APIs share packed model state, release the GIL and bound their query workers by the selected CPU allocation; `threads=0` uses that allocation.

Independent checks. `concordance.py` compares native nucleotide and amino-acid scores to OLGA on the same legacy parameter family and checks synonymous-codon sums when a sufficiently short generated query is available. It is an engine check, not validation of an independently fitted model. `compare_models.py` separates this raw-reference oracle comparison from software validation on freshly annotated functional-source reads outside the nonfunctional training inputs. Functional-source reads have undergone selection, so their likelihood is not an unbiased held-out recombination objective. The accepted validator selects the first 20000 raw reads and then the first 300 eligible unique junctions per context; the source bucket alone does not prove every selected junction is productive. For its OLGA oracle, three zero insertion bins extend the iteration workspace without changing probability coefficients. Focused tests cover joint-history normalisation, genomic constraints, allele aliases, retained-germline errors and pruning, as well as the independent validator. The historical timing and entropy tables above were not rerun for this revision.

Scope limits. The present engine supports the declared VJ, single-D and up-to-two-D factorisations, not arbitrary Bayesian-network structures. The observation channel covers retained-germline substitutions, not indels or errors on inserted bases. Candidate-gene restriction and positive inference cutoffs can remove compatible histories; the latter reports actual omitted mass but does not bound it by the cutoff. Gene-level reference selection does not infer an individual's allele genotype. The entropy diagnostics describe the declared parameter support; dinucleotide marginal entropies use stationary composition, and scenario-entropy contributions weight first-base, transition and tandem events by their activation probabilities. These summaries do not replace independent validation on matched references and inputs.

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